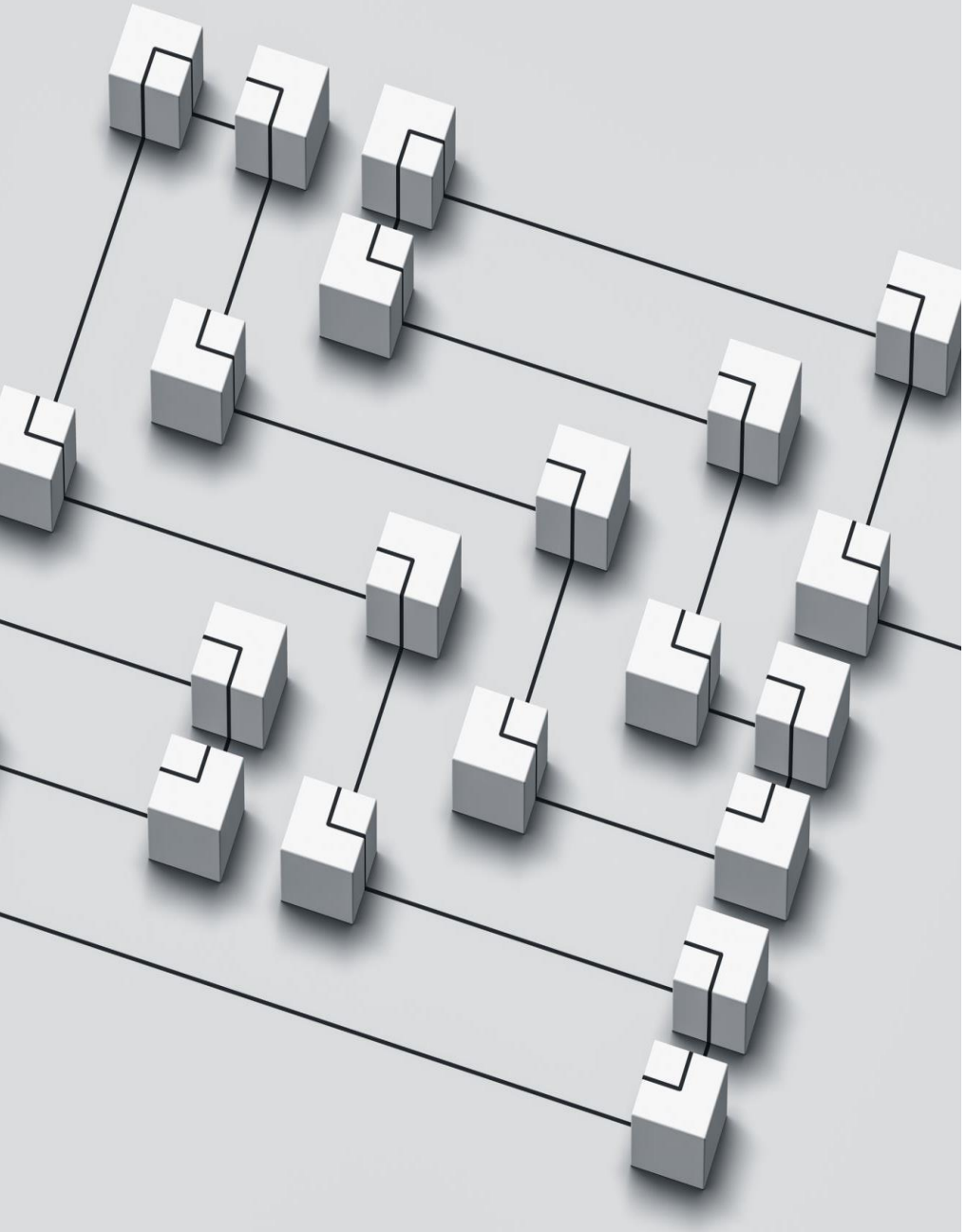
COM4401 Foundations of Computer Science and Academic Skills

Team Portfolio Presentation

Members : Seerat, Aiman, Jimmy, Faris



Portfolio Overview

This portfolio consists of four mini-projects covering key areas of computer science:

Mini Project 1:

- Workstation Design and Database Development

Mini Project 2:

- Pizza Ordering System

Mini Project 3:

- Network Design and Configuration

Mini Project 4:

- Cyber Security Awareness

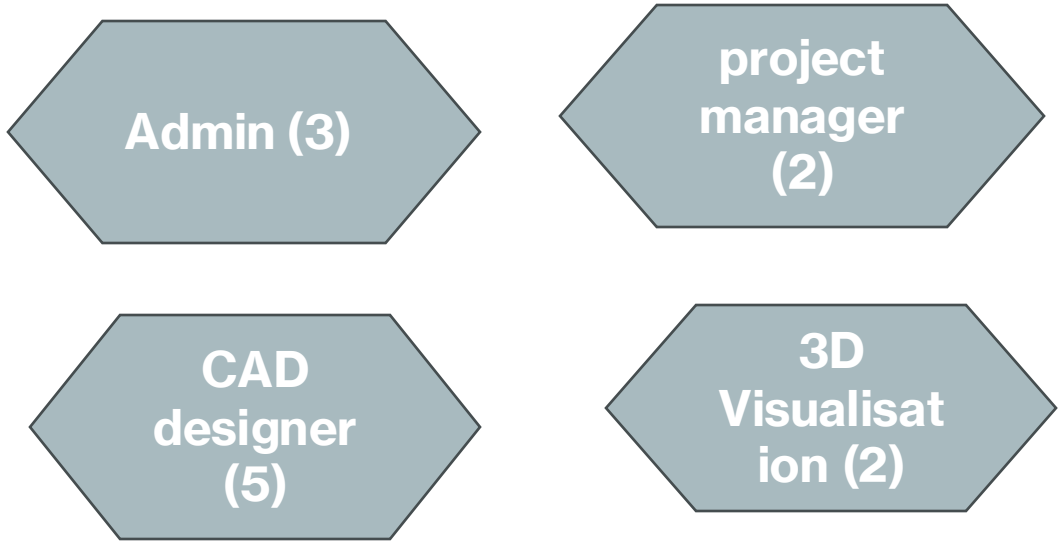
Objectives:

- Apply computing theory to practical scenarios
- Work collaboratively as a team
- Develop technical and problem-solving skills
- Produce professional documentation and presentations

Mini project 1- Northwest Architecture Studio Workstations

Northwest Architecture Studio LTD requires reliable workstations across 4 different employee roles. Our aim is to recommend suitable computer hardware for each of the roles while keeping costs within the budget that was allocated .

12 workstations. 4 different roles



Different employees require different levels of computer performance.

Admin staff need affordable and reliable workstations.

Project managers need fast multitasking and efficient communication.

CAD and 3D specialists require powerful hardware for modelling and rendering.

Balance performance reliability and cost within the £26,000 maximum budget.

Each workstation must include suitable monitors network and connectivity for daily work.

BUDGET	AMOUNT
Hardware Budget	£24,000
Contingency	£2,000
Total available	£26,000

Our Solution – recommended workstations

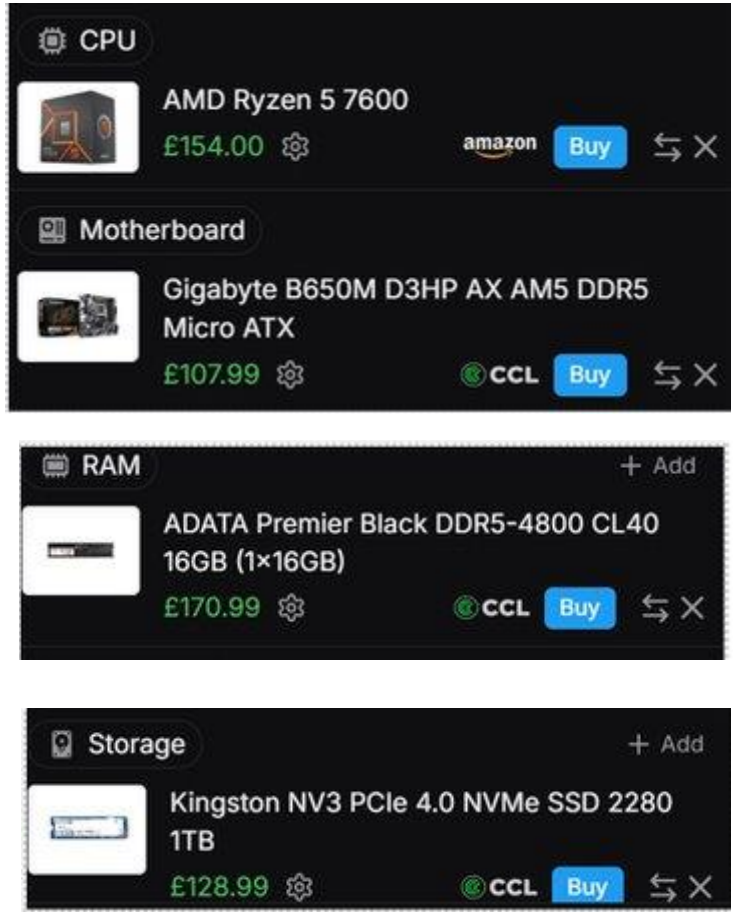
Why our solution works

- Hardware matched to each employee's responsibilities.
- Dedicated GPUs support CAD modelling and 3D rendering.
- Faster SSDs and more RAM improve productivity.
- Different specifications avoid unnecessary spending.
- Integrated graphics save money on the office workstations
- We compared prices and compatibility to recommend suitable workstations within the overall £26,000 maximum budget.
- Our workstation choices allow for future upgrades as the company needs to grow

Role	CPU	Ram/storage	Why chosen?
Admin (3)	Ryzen 5 5600G	16GB DDR4, 512GB SSD	Affordable for office tasks
PROJECT Manager (2)	Ryzen 5 7600	16GB DDR5, 1TB SSD	Fast multitasking and meetings
CAD designer (5)	Intel core i7-13700k	32GB DDR5, 1TB SSD	Technical drawings and modelling
3D Visualisation (2)	AMD Ryzen 9 7900X	64GB DDR5, 2TB SSD	Demanding rendering tasks

Our selection process

- Researched suitable hardware for each job role
- Compared component prices and performance.
- Checked CPU motherboard and ram compatibility.
- Selected components which are reliable and within budget.
- We documented our final recommendations in the SQL database.



Final results and Budget evaluation

- The Admin computers cost £1,971.
- The CAD computers cost £13,880, which was over their budget.
- The 3D vis computers cost £6,873 which were under the budget.
- Altogether, the total was £25,178.
- This was within our overall £26,000 limit, including contingency.

Role	Budget	Actual cost
Admin (3)	**£2550**	**£1,971.09**
Project manager (2)	**£2500**	**£2,453.88**
CAD designer(5)	**£11,750**	**£13,880.10**
3d visualisation(2)	**£7,200**	**£6,873.40**
Total	**£24,000**	**£25,178.47**

Mini Project 2 - Pennine Pizza Works

Problem

Manual order calculations

Staff using paper and calculators

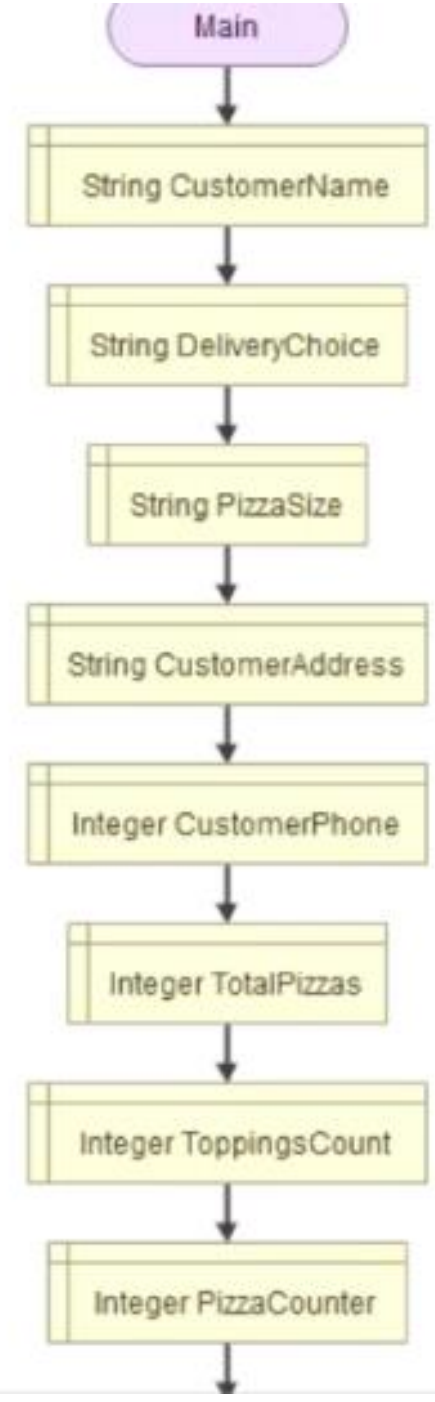
Pricing errors

Solution

Flowgorithm ordering system

Automated calculations

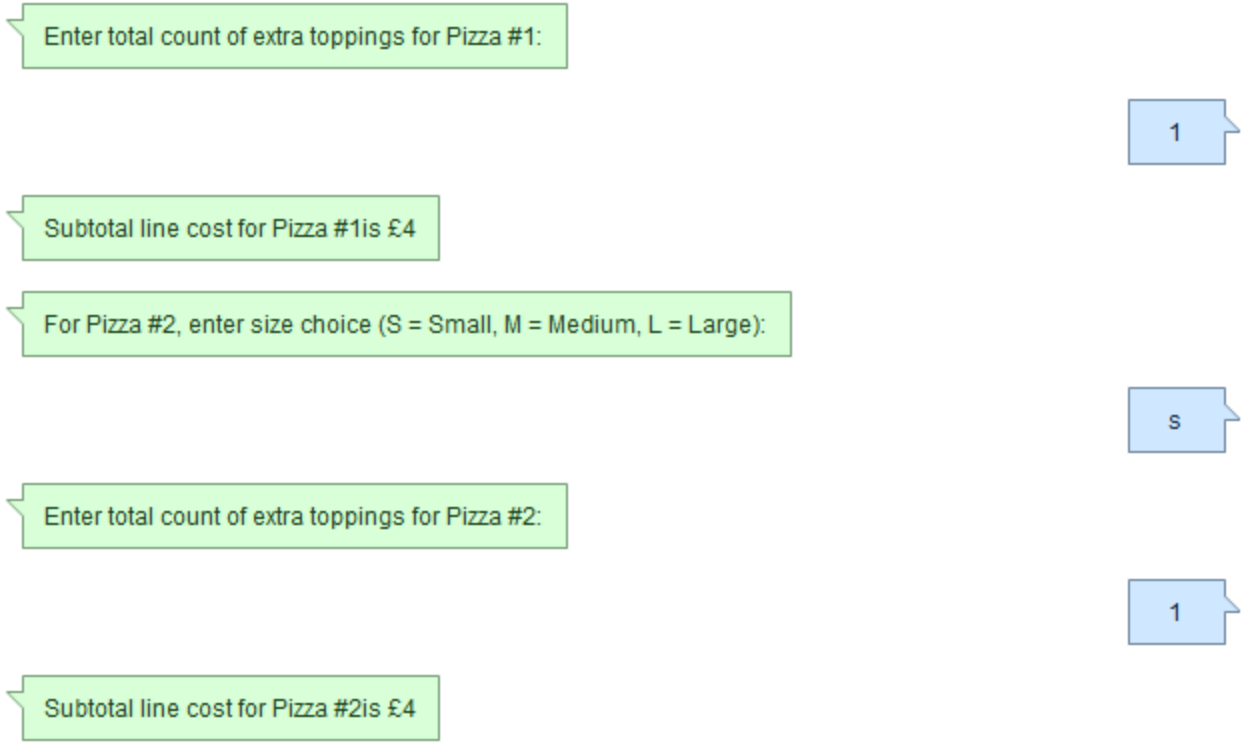
Itemised customer bills



Issue Found

Decimal values removed from prices

Example: £3.25 became £3



Fix

Changed Integer variables to Real variables

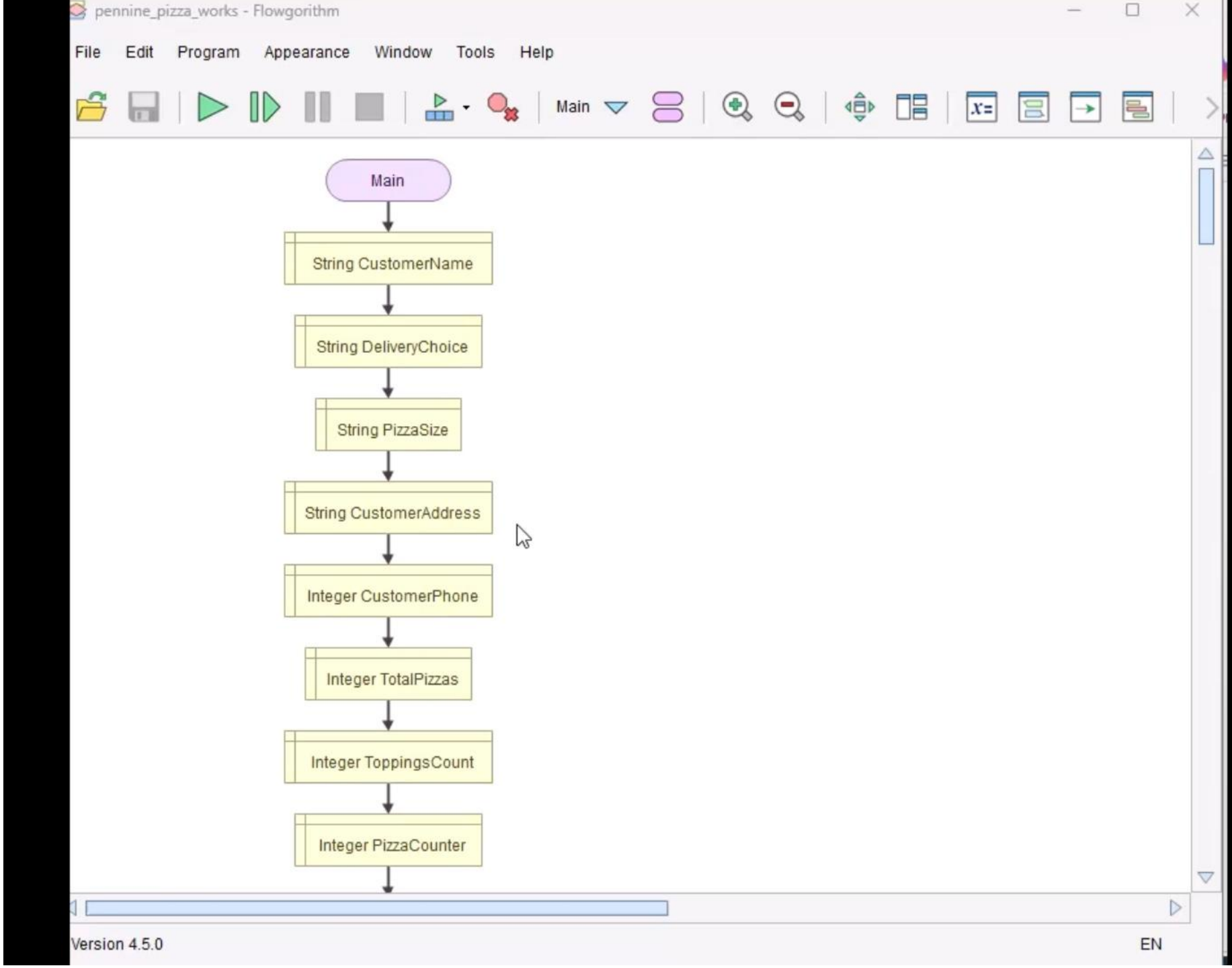
Result

Accurate pricing calculations

Test Table

Test Category	Inputs	Type	Outcome	Pass/fail
Pizza Qty	0,1,6,7	Extreme / Abnormal	Accepted 1 & 6 Blocked 0 & 7	Pass
Data Type	“Three” , 1.5	Abnormal	System rejected non-integer types	Pass
Menu Sizing	S , L , M , X , “ “	Normal / Abnormal	Handled case - insensitivity Blocked X / blank	Pass
Topping Counts	0, 4, 5	Normal / Extreme	Charged £0.00 Correctly capped 4+ at £2.50	Pass
Delivery Flag	y, N,	Normal / Abnormal	Accepted variations of Y/N Rejected text	pass
Discount Treeshold	£20.00, £20.01	Extreme	£20.00 gave no discount £20.01 gave 10%	Pass

Flowgorithm Demo



Mini Project 3 – Northwest Architecture Studio Network

We designed and built a LAN in Filius for Northwest Architecture Studio.

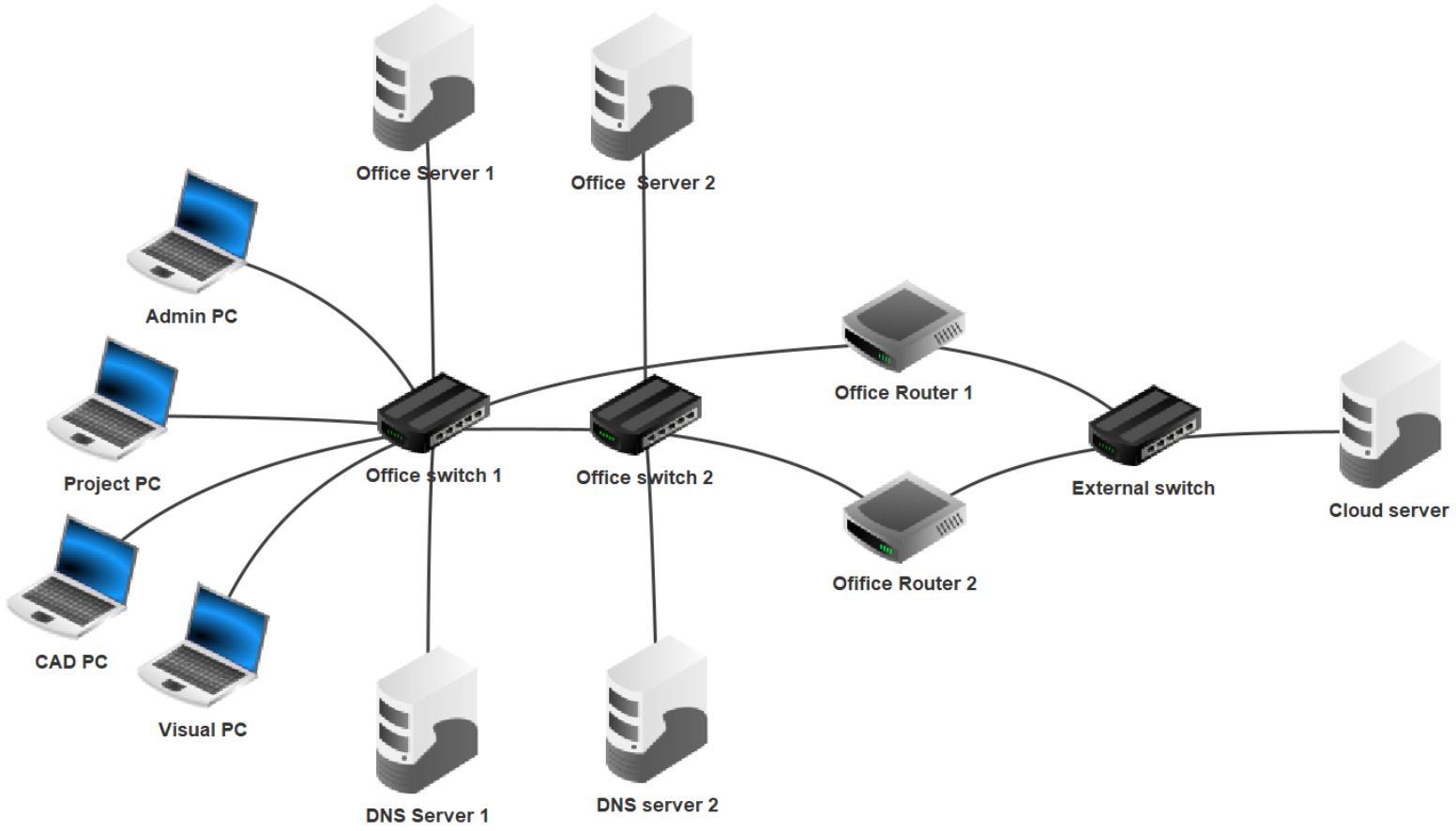
Office LAN: 192.168.10.0/24

External networks: 203.0.113.0/24

Routers act as the gateway between the networks

DNS turns names into IP addresses

Redundancy : 2 routers , 2 switches , 2 DNS and 2 servers

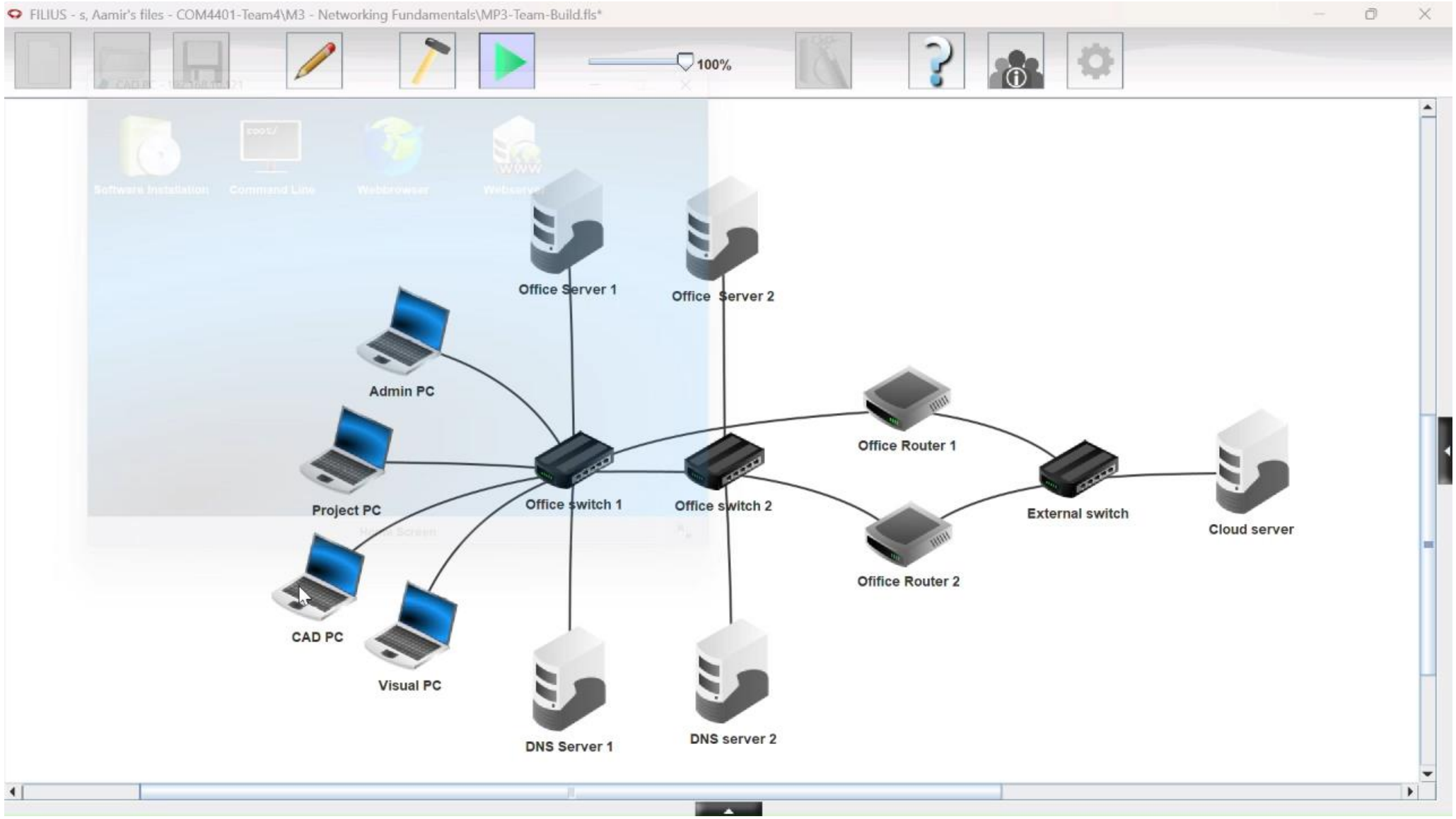


Test Table

We tested the network from the CAD PC. All pings were successful with 0% packet loss. However. The TTL value decreased from 64 to 63 when pinging the Cloud Server. Which shows the packets passed through one router.

Test (from CAD PC)	Command	Result
PC to Admin PC	Ping 192.168.10.101	Pass
PC to Project PC	Ping 192.168.10.111	Pass
PC to Visual PC	Ping 192.168.10.131	Pass
PC to Office Server	Ping 192.168.10.10	Pass
PC to DNS Server	Ping 192.168.10.2	Pass
PC to Cloud Server	Ping 203.0.113.10	Pass (TTL 63)

Test Video


FILIUS - s, Aamir's files - COM4401-Team4\M3 - Networking Fundamentals\MP3-Team-Build.fls*
100%
Office Server 1
Office Server 2
Admin PC
Project PC
CAD PC
Visual PC
Office switch 1
Office switch 2
Office Router 1
Ofifice Router 2
External switch
Cloud server
DNS Server 1
DNS server 2

Redundancy Testing

Demo 1: Office Server failure

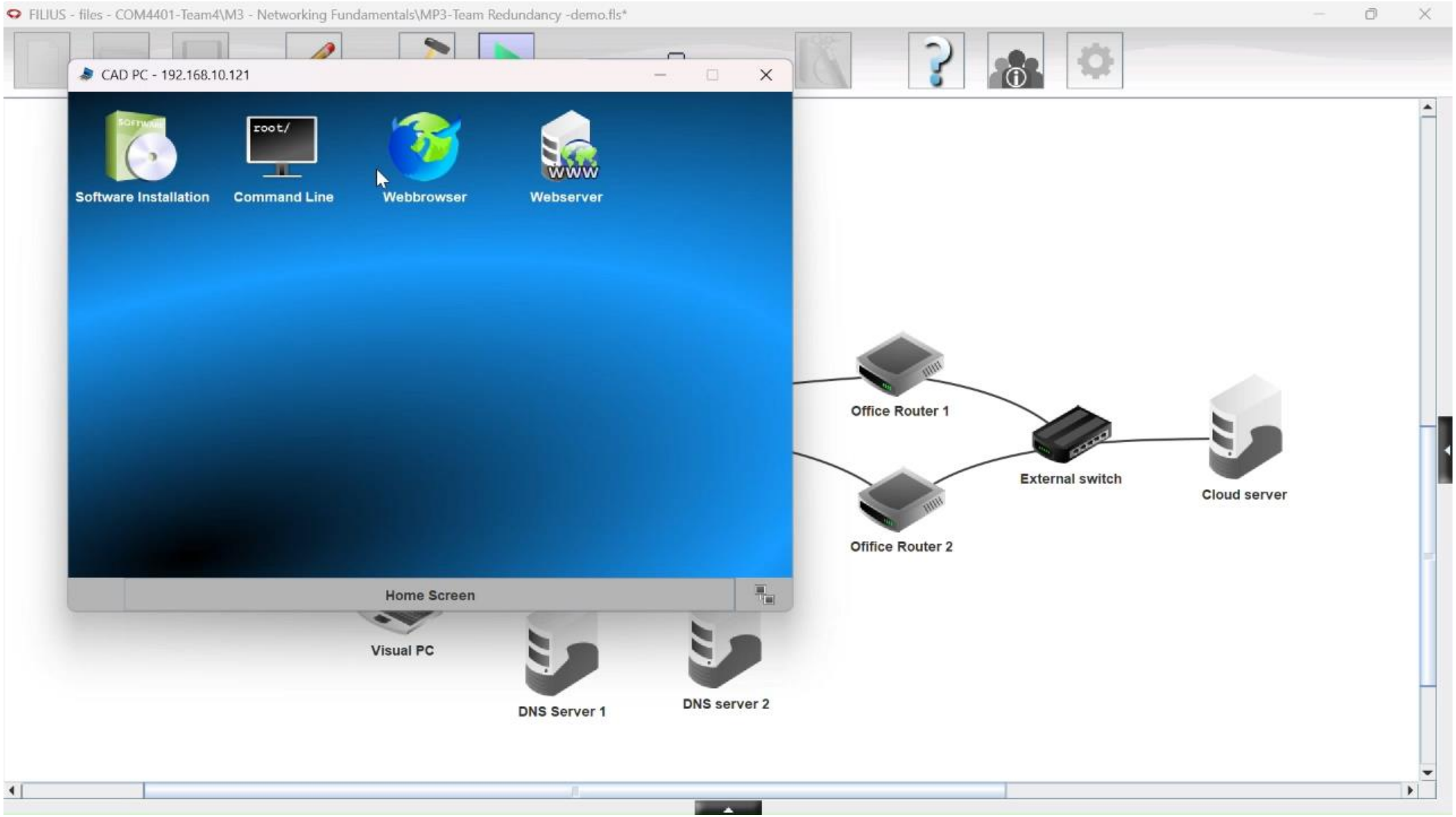
- Office Server 1 was disconnected, so the intranet showed "server does not exist"
- The DNS record was changed to 192.168.10.11, so the backup page loaded from Server 2

Demo 2: DBS Server failure

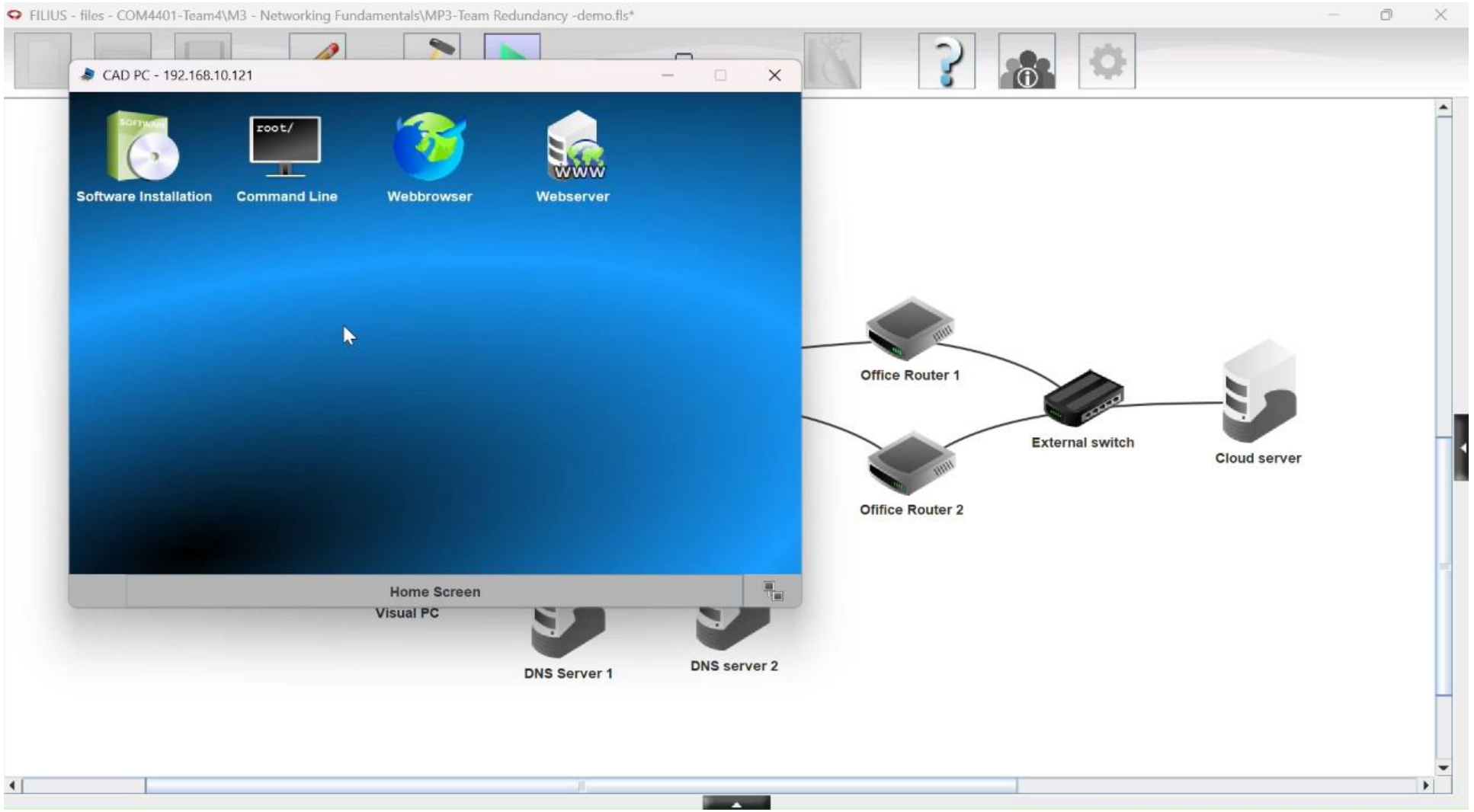
- DNS Server 1 was disconnected: the intranet failed by name but still worked by IP
- The CAD PC'S DNS was changed to 192.168.10.3, so the name worked again through the DNS Server 2

Limitation: the failover was manual; automatic failover would be better.

Demo 1 Office Server failure



Demo 2 DNS Server failure



Mini Project 4 – Cyber Security Awareness

Project Aim

To educate users about common cyber threats and how to protect themselves online.

Themes Covered

- Password Hygiene
- Multi-Factor Authentication
- Phishing and Social Engineering
- Safe Device Usage
- Public Wi-Fi Security

Importance

Cyber-attacks increasingly target human behaviour rather than technical vulnerabilities.

What is Phishing?

A phishing attack is a form of cybercrime where attackers attempt to trick users into revealing sensitive information.

Common Types

- Email Phishing
- Spear Phishing
- Smishing (SMS)
- Vishing (Voice Calls)

Targets

- Passwords
- Banking details
- Personal information
- Company credentials

Recognising Phishing Attacks



Warning Signs of Phishing



Red Flags:

Urgent requests
Suspicious email addresses
Unexpected attachments
Requests for passwords
Poor grammar and spelling



Prevention Measures:

Verify the sender
Check website URLs
Hover over links before clicking
Enable Multi-Factor Authentication
Report suspicious emails

Key Lessons Learned

Skills Developed Across All Projects

Computing Fundamentals	• Hardware selection • Database design
Software Engineering	• Requirements analysis • Program design • Testing
Networking	• Network configuration • DNS implementation • Troubleshooting
Cyber Security	• Threat awareness • Risk identification • User protection strategies

Conclusion

Key Takeaways
Practical experience across multiple computing disciplines
Development of technical and professional skills
Improved teamwork and communication
Better understanding of real-world computing problems

Thank you for listening

Any Questions?